

BALANCE-FIELD FRAMEWORK (BFG)

Canonical Minimal Finite Closure and Universal Reclosure

A Preprint on Typed-Operational Minimality, Neutral Selection, Intrinsic-Gram Reconstruction, Identity Continuity, and Structural Emergence

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Scientific status.

Independent mathematical preprint and falsifiable research program. The finite operator identities and theorem statements are mathematical consequences only under their stated assumptions. The present work does not establish the Balance-Field Framework as empirically confirmed fundamental physics or as a completed Theory of Everything. It isolates the current minimal finite BFG closure core and the exact or source-synthesized operations used by the living reclosure candidate.

Source discipline.

No external physical theory is used as a constitutive premise of the finite construction. BFG source documents determine the architecture. External mathematical literature is cited only for standard mathematical tools and established background results; external physical literature appears only in comparison and recovery discussions.

Abstract

This preprint consolidates the current finite Balance-Field Framework (BFG) closure program into a single minimal and auditable formulation. The operative state is the five-object quotient $\Xi=(K,\rho_F,\rho_{W,Y},P)$, comprising Hermitian formation geometry, a nonzero positive formation/load density, a normalized positive identity witness, a positive neutral load, and an orthogonal persistent projector. Within the present finite candidate this state is typed-operationally minimal: deleting any one role can make the successor indeterminate. The universal neutral pair $C_N=(I+Y)^{-1}$ and $B_N=Y(I+Y)^{-1}$ is exact and complementary. A source-synthesized Neutral-Contrast Selection identifies closure-positive modes through $Q_N=1_{(0,\infty)}[(I-Y)(I+Y)^{-1}]$, without a fitted threshold. Selection precedes generation of the next packet. The surviving source determines reciprocal closure loads and a parameter-free dual weight pair; these define a cross-fed analysis operator whose reduced singular values generate the successor positive geometry $Y_+=\Sigma^2$. The same active frame transports the formation operator, while a distinct R4 polar overlap carries inherited identity and formation into the successor without relabelling emergent formation as prior identity. A simple negative ground mode in the non-inherited target complement supplies the only admissible endogenous formation seed. The resulting map is a level-independent finite self-reclosure law with no free numerical or constitutive completion function in the current minimal kernel, subject to the explicitly stated source-synthesis boundary. An exact BFG inheritance-emergence theorem further shows, on the neutral-compatible persistent stratum, that $[Y_P,K_P]=0$ is equivalent to formation inheritance, whereas

noncommutation forces a change in the formation spectrum. Numerical audits reported in the BFG source line test the finite identities and novelty mechanism to floating-point precision but are not empirical evidence for natural realization. The preprint closes by locating continuum phase, memory, minimal polar carrier geometry, gravity, quantum structure, and cosmology as downstream derivation obligations rather than new fundamental inputs.

Keywords: Balance-Field Framework; recursive closure; neutral mediation; operator theory; polar decomposition; singular value decomposition; persistence; identity witness; structural emergence; no-retuning; minimal state.

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1. Purpose, scope, and claim discipline

The purpose of this preprint is to place the present finite BFG research program in one internally ordered mathematical document. The target is not to add another layer of mechanisms. The target is to identify the smallest currently sufficient finite state, derive the successor from already present BFG roles, and mark every remaining assumption or branch condition explicitly. This consolidates the natural-derivation methodology of the living BFG package [3–8].

Existence = stable admissible recursive persistence (1)

The finite mathematical content is therefore organized around a single level-independent state transformation rather than around separate domain laws. Physical, biological, cognitive, or phenomenological realizations are downstream interpretation problems.

$$\Xi_{n+1} = \mathcal{U}_{\text{BFG}}(\Xi_n), \quad \mathcal{U}_{\text{BFG}}(\perp) = \perp \quad (2)$$

Table 1. Epistemic discipline used throughout the preprint.

Claim class	Meaning in this preprint
Definition	Introduces a state object, operator, equivalence, or terminal state.
Theorem	Exact consequence of the stated assumptions.
Source-synthesized construction	Combines BFG-explicit roles by a standard mathematical operation without adding a free parameter.
Conditional theorem	Exact consequence after an explicitly stated branch assumption.
Numerical audit	Finite-precision stress test of mathematics or implementation; not empirical validation.

Claim class	Meaning in this preprint
Empirical realization	A future claim that measured natural systems instantiate the frozen BFG law.

2. Source discipline and mathematical background

The constitutive source line is BFG-internal. In particular, the minimal finite state, the neutral pair, the two-density separation, Selection-first ordering, reciprocal dual balance, intrinsic-Gram successor load, formation transport, and R4 identity continuation are taken from or synthesized within the BFG research corpus [1–8].

Standard matrix and operator theory is used only as mathematical language: spectral functional calculus, orthogonal and metric projections, SVD, polar decomposition, Moore–Penrose inversion, positive Gram operators, and perturbation estimates are established tools [16–30]. Their use does not make the BFG law a patchwork of external physical theories.

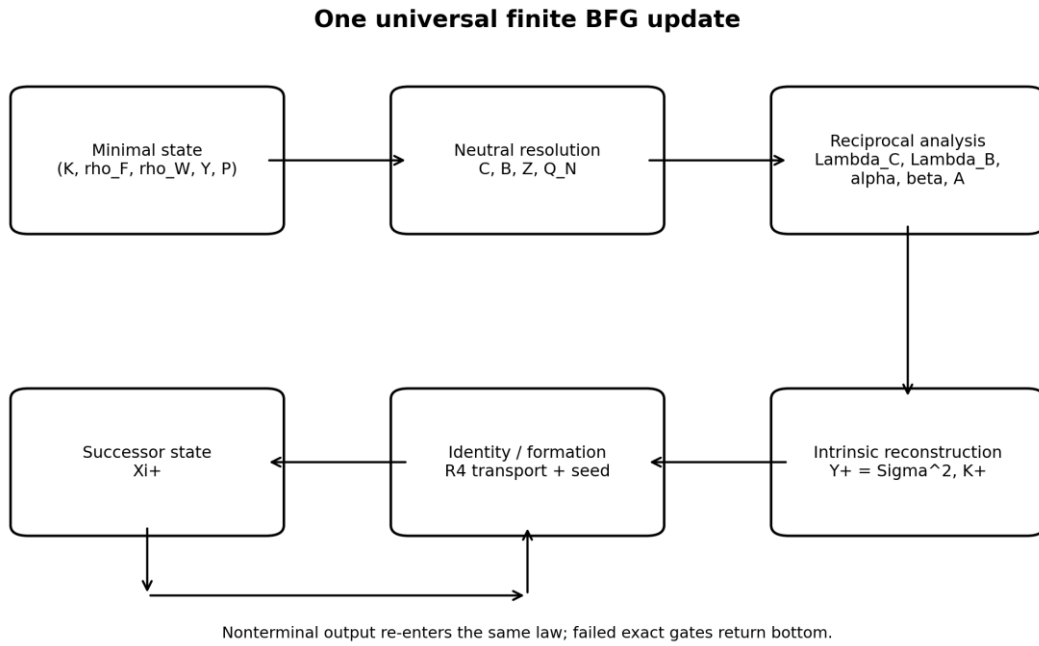


Figure 1. Canonical organization of the finite BFG update. The boxes are not separate fundamental laws; they are deterministic stages of a single state map $\mathcal{U}_{\text{BFG}}$. Failed exact gates return the absorbing terminal state $\perp$.

3. The minimal finite BFG state

The current finite closure dynamics factors through a five-object quotient. The common finite carrier is implicit in matrix dimension and is therefore not an additional dynamic coordinate [3,4].

$$\mathbb{E}_{\min} = (K, \rho_F, \rho_W, Y, P) \tag{3}$$

$$K = K^\dagger \tag{4}$$

$$\rho_F \geq 0, \quad \text{tr } \rho_F > 0 \tag{5}$$

$$\rho_W \geq 0, \quad \text{tr } \rho_W = 1 \tag{6}$$

$$Y > 0 \tag{7}$$

$$P = P^\dagger = P^2 \tag{8}$$

The five roles are typed rather than merely stored: K carries formation/balance; ρ_F carries present formation/load distribution; ρ_W carries inherited identity continuity; Y carries neutral mediation; P carries persistent support. The BFG minimality audit provides counterexamples showing that no one role is universally recoverable from the remaining four [4].

Table 2. Typed-operational roles of the minimal finite state [4].

Object	Role	Why it cannot be deleted in the current finite candidate
K	Formation / balance geometry	Changing K alone can switch an emergent block between a valid simple negative seed and terminal nonformation.
ρ_F	Formation / load density	Changing ρ_F changes reciprocal loads, weights, singular values, and therefore Y_+ .
ρ_W	Identity witness	Different witnesses on the same K, ρ_F, Y, P can survive or be annihilated by the same Selection event.
Y	Neutral geometry	Changing Y changes C_N, B_N, Z , Selection, graph metric, reciprocal loads, and the successor Gram.
P	Persistent support	States with identical K, ρ_F, ρ_W, Y but different persistent rank can have different successors.

4. Exact neutral resolution

Let Y be a positive neutral load. The exact BFG neutral resolution is the complementary operator pair already present in the Universal Structural and Canonical Reclosure lines [1,11].

$$C_N(Y) = (I + Y)^{-1} \quad (9)$$

$$B_N(Y) = Y(I + Y)^{-1} = I - C_N(Y) \quad (10)$$

$$C_N(Y) + B_N(Y) = I \quad (11)$$

$$G(Y) = I + Y \quad (12)$$

$$Z(Y) = C_N(Y) - B_N(Y) = (I - Y)(I + Y)^{-1} \quad (13)$$

Because C_N and B_N are complementary functions of the same positive operator, the neutral split introduces no independent response law. The exact contrast Z is a lossless encoding of the pair.

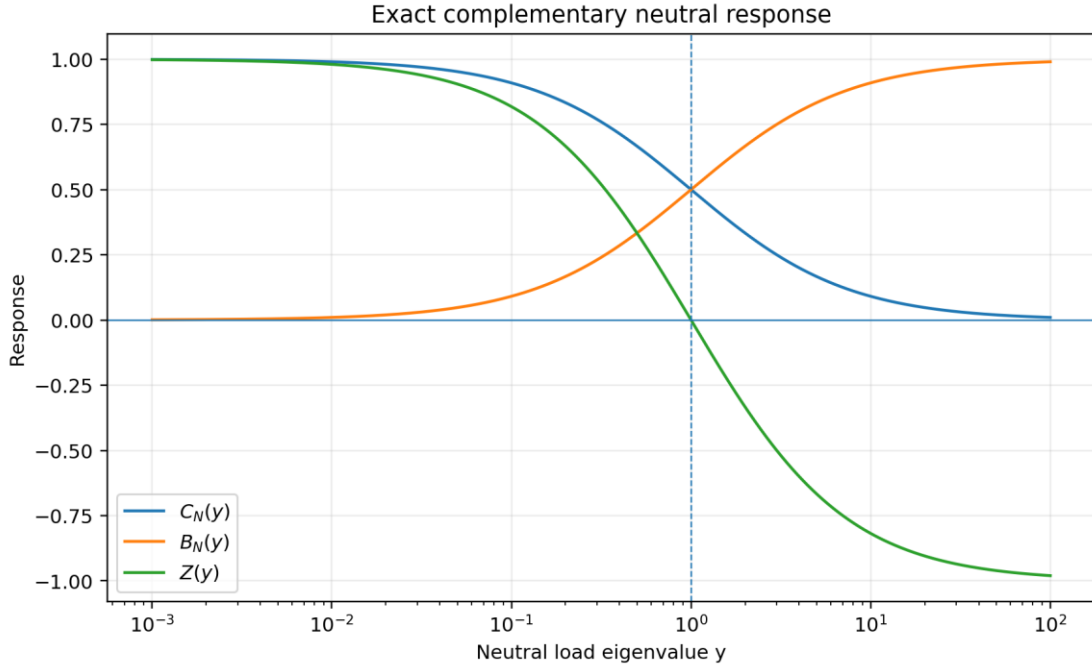


Figure 2. Exact neutral response on a scalar spectral mode. The crossing $y=1$ is the intrinsic role-balance point $C_N=B_N$, not an externally fitted threshold.

Theorem 1 (Exact neutral complementarity). For $Y \geq 0$, C_N is positive with $0 < C_N \leq I$, B_N is positive with $0 \leq B_N < I$, and $C_N + B_N = I$. The contrast Z has spectrum in $[-1, 1]$.

Proof sketch. Apply scalar functional calculus to $y \geq 0$ and lift the resulting inequalities spectrally [16–19].

5. Selection-first neutral admissibility

The current finite ordering applies Selection to the present closure before generating the next packet. This follows the BFG source ordering $A \rightarrow S_A \rightarrow T_A$ and avoids a built-in bias toward preserving every previously persistent direction [8].

$$Q_N = \mathbb{1}_{(0, \infty)}(Z) \quad (14)$$

$$Q_N = \mathbb{1}_{[0, 1)}(Y) \quad (15)$$

The second equality follows because $z(y) = (1-y)/(1+y)$. Exact equality $y=1$ is assigned to export; floating arithmetic near $y=1$ is reported as numerical ambiguity rather than converted into a physical tolerance.

$$T_{\text{sel}} = \text{polar}(Q_N P) \quad (16)$$

T_{sel} is the canonical partial-isometric survival transport from the old persistent support into the currently closure-positive source sector. Formation is transported without normalization, while the identity witness remains normalized after retained-mass accounting.

Theorem 2 (Parameter-free neutral Selection). Under the source-synthesized identification of closure-positive neutral mode with retained response exceeding complementary response, the universal selector is $Q_N = \mathbb{1}_{(0, \infty)}(Z)$, with no fitted threshold, gain magnitude, or extra response channel [6,8].

Epistemic boundary. The retained/complementary neutral roles and their exact balance surface are source-explicit; their use as the universal finite Selection boundary is a cross-source BFG synthesis rather than a single historical equation [6].

6. Reciprocal dual balance and the cross-fed analysis map

Let $P^\wedge(G)$ denote the G -orthogonal projector onto the selected persistent source. The selected formation density determines two complementary closure loads [3,7].

$$\Lambda_C = \text{tr}(G P^\wedge(G) C_N \rho_F C_N P^\wedge(G)^\dagger) \quad (17)$$

$$\Lambda_B = \text{tr}(G P^\wedge(G) B_N \rho_F B_N P^\wedge(G)^\dagger) \quad (18)$$

$$\alpha = \Lambda_B / (\Lambda_C + \Lambda_B), \quad \beta = \Lambda_C / (\Lambda_C + \Lambda_B) \quad (19)$$

$$\alpha + \beta = 1, \quad \alpha \Lambda_C = \beta \Lambda_B \quad (20)$$

Thus reciprocal order is not an externally fitted pair. For strictly positive dual loads it is uniquely fixed by the state.

$$\mathcal{A} = [\sqrt{\alpha} P^\wedge(G) C_N ; \sqrt{\beta} P^\wedge(G) B_N] \quad (21)$$

The two neutral channels therefore produce a single selected cross-fed analysis map. The arithmetic-geometric-mean estimate in the Canonical Universal Reclosure architecture shows that this construction is non-amplifying in the structural graph metric [1].

7. Intrinsic-Gram successor geometry

The successor neutral load is generated by the transformation that actually occurred. Let the reduced singular value decomposition of the selected analysis map be

$$\mathcal{A} = U \Sigma V^\dagger \quad (22)$$

$$\mathcal{A}^\dagger \mathcal{A} = V \Sigma^2 V^\dagger \quad (23)$$

The positive quadratic form $q(x) = \|\mathcal{A}x\|^2$ has the unique positive representing operator $\mathcal{A}^\dagger \mathcal{A}$ on the active source support. In reduced active coordinates the successor load is therefore

$$Y_+ = \Sigma^2 \quad (24)$$

Theorem 3 (Intrinsic-Gram successor load). Once the selected analysis map $\mathcal{A}$ is fixed, its induced positive quadratic geometry is unique. In the reduced singular-vector coordinates this geometry is Σ^2 ; hence no independent next-load coefficient or response function is required [3,5].

This theorem is a BFG derivation using standard matrix representation theory. It should be distinguished from the historical question of whether every earlier BFG paper had already written $Y_+ = \Sigma^2$ explicitly.

Intrinsic-Gram successor surface (retained scalar stratum)

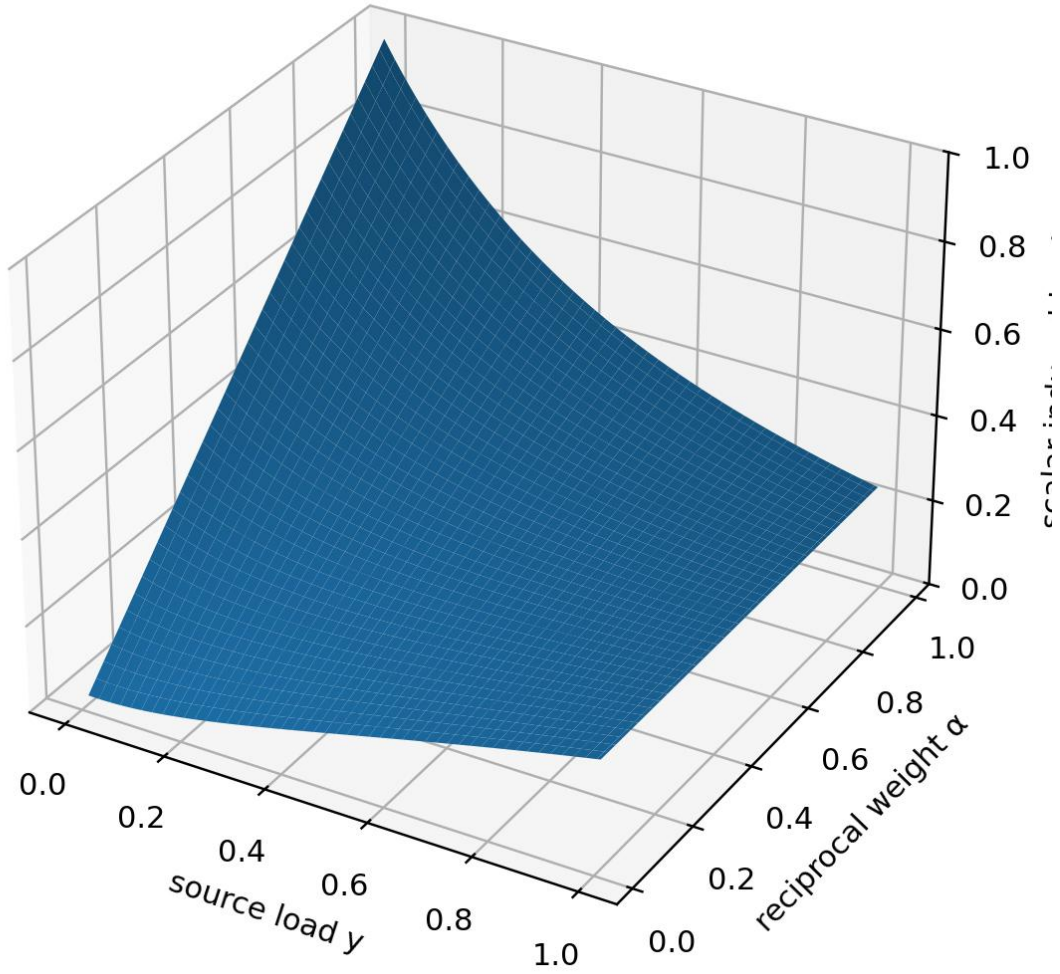


Figure 3. Scalar visualization of the induced successor load $y_+ = [\alpha + (1-\alpha)y^2] / (1+y)^2$ on a retained one-mode stratum. The surface is a mathematical visualization, not empirical data.

8. Formation transport, identity continuity, and emergent seeding

Because the rich BFG capacity operators share one linear transport, their dynamically visible finite combination factors exactly through the Hermitian formation operator K [3,4].

$$K = \mathfrak{c} + \mathfrak{K} - \mathfrak{D} \quad (25)$$

$$K_+ = U^\dagger (K \oplus K) U \quad (26)$$

The two positive state densities must remain distinct. ρ_F carries current formation/load magnitude; ρ_W is a normalized witness of inherited identity. A single untagged positive operator cannot satisfy both roles across a seeded rank increase [7].

$$S_- = T_{R4}^\dagger T_{R4}, \quad S_+ = T_{R4} T_{R4}^\dagger \quad (27)$$

$$m_{\text{keep}} = \text{tr}(S_- \rho_W) \quad (28)$$

$$\rho_{W,+} = T_{R4} \rho_W T_{R4}^\dagger / m_{\text{keep}} \quad (m_{\text{keep}} > 0) \quad (29)$$

$$\rho_F^{\text{inh}} = T_{R4} \rho_F T_{R4}^\dagger \quad (30)$$

Let $E = I - S_+$ be the active target complement not inherited from the predecessor. If $E \neq 0$, a new formation seed is admitted only when the compressed operator K_E has a simple negative lowest eigenvalue.

$$K_E = E K_+ E \mid_{\neg} (\text{Ran } E) \quad (31)$$

$$\rho_E = (-\lambda_0^E) \Pi_E \quad (32)$$

$$\rho_{F,+} = \rho_F^{\text{inh}} + \rho_E \quad (33)$$

$$P_+ = S_+ + \Pi_E \quad (34)$$

No emergent formation is relabelled as inherited identity: $E\rho_{W,+}=0$ by construction. This is the central reason for the explicit two-density state.

9. The canonical finite master reclosure law

The complete finite update can now be stated as one map. Its internal stages are deterministic consequences of the current state, standard matrix operations, and exact no-choice gates.

$$\mathcal{U}_{\text{BFG}} : (K, \rho_F, \rho_W, Y, P) \mapsto (K_+, \rho_{F,+}, \rho_{W,+}, Y_+, P_+) \quad (35)$$

$$\Xi_{n+1} = \mathcal{U}_{\text{BFG}}(\Xi_n) \quad (36)$$

For the richer semantic/source representation, a canonical transverse recursion representative may be reconstructed after the minimal update:

$$R_{C,+} = P_+ + (I - P_+) C_N(Y_+) (I - P_+) \quad (37)$$

This representative is not an additional finite constitutive law. Once P_+ and Y_+ are known, it is a derived rich-state lift [3,4].

Table 3. Canonical finite BFG update pipeline.

Stage	Input role	Canonical operation	Output
1	Y	Exact neutral functional calculus	C_N, B_N, G, Z
2	Y,P	Neutral-Contrast Selection + polar survival	Q_N, T_{sel}
3	ρ_F, Y, P	Reciprocal load traces	$\Lambda_C, \Lambda_B, \alpha, \beta$
4	Selected source	Cross-fed analysis	$\mathcal{A}$
5	$\mathcal{A}$	Reduced SVD / positive Gram	U, Σ, V and Y_+
6	K,U	Common linear transport	K_+
7	ρ_W, ρ_F	R4 polar overlap	Inherited witness and formation
8	K_+ on emergent complement	Simple-negative-ground no-choice seed	ρ_E, Π_E
9	Inherited + emergent sectors	State commit	Ξ_+ or $\perp$

Theorem 4 (Finite BFG master reclosure, current candidate). For an admissible minimal state Ξ , the Selection-first finite construction determines either a unique nonterminal successor Ξ_+ of the same five-object type or the absorbing terminal state $\perp$, without a free numerical or constitutive completion function in the minimal kernel. Under a conservative single-document audit, one cross-source role identification remains visible: the Neutral-Contrast operationalization of closure-positive Selection [3,6].

10. Typed-operational minimality

Minimality is asserted here in a typed operational sense, not in the set-theoretic sense that several matrices cannot be encoded into a single symbol. The current finite update requires five independent information roles [4].

Theorem 5 (Typed-operational minimality). Within the current finite candidate and declared state types, deleting any one coordinate from $(K, \rho_F, \rho_W, Y, P)$ destroys deterministic successor evolution on some admissible state family. Therefore the five-object quotient is typed-operationally minimal [4].

The theorem does not declare the richer capacity triple, recursive operator, or semantic BFG variables meaningless. It states only that they are not required to compute the present finite successor once their dynamically visible quotient information is known.

11. Inheritance, reproduction, and structural emergence

A nontrivial state transition is not automatically emergence. Canonical equivalence to the inherited active-support reference is persistence or reproduction; inequivalence is structural novelty. On the neutral-compatible persistent stratum the BFG reclosure admits an exact closed form [2].

$$\chi_{ij} = (\alpha + \beta y_i y_j) / \sqrt{[(\alpha + \beta y_i^2)(\alpha + \beta y_j^2)]} \quad (38)$$

$$(K_+)_ij = \chi_{ij} (K_P)_ij \quad (39)$$

$$\text{tr}(K_P^2) - \text{tr}(K_+^2) = \alpha\beta \sum_{ij} [(y_i - y_j)^2 |(K_P)_{ij}|^2] / [(\alpha + \beta y_i^2)(\alpha + \beta y_j^2)] \quad (40)$$

Theorem 6 (Inheritance-emergence dichotomy on the neutral-compatible stratum). $[Y_P, K_P] = 0$ if and only if $K_+ = K_P$. If $[Y_P, K_P] \neq 0$, then $\text{tr}(K_P^2) > \text{tr}(K_+^2)$, hence the formation spectra differ. The relevant noncommutativity is therefore internal mismatch between inherited formation and neutral geometry, not merely $[K, P]$ [2].

Exact off-block reclosure attenuation for $\alpha = \beta = 1/2$

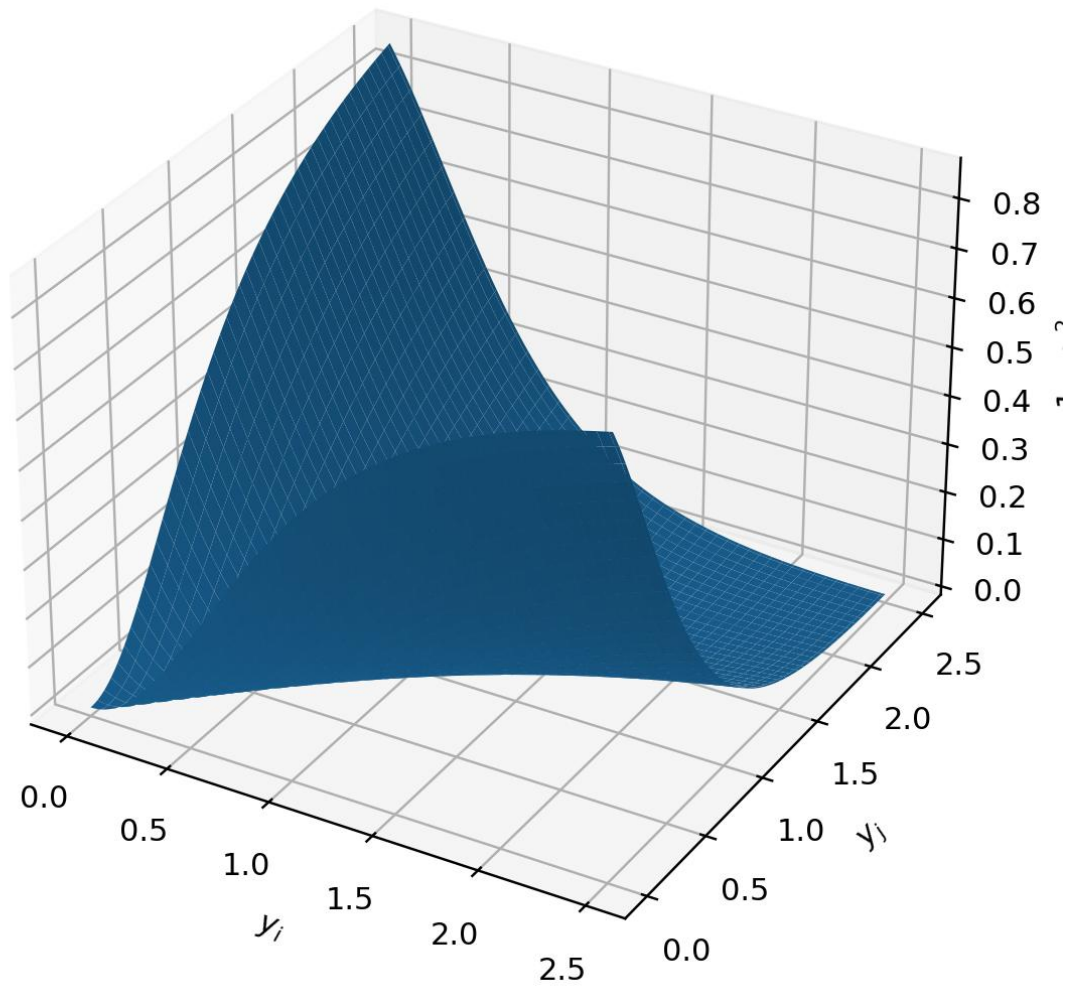


Figure 4. Exact reclosure attenuation $1 - \chi^2$ for a pair of neutral eigenvalues at $\alpha = \beta = 1/2$. The attenuation vanishes on $y_i = y_j$ and is strictly positive across distinct neutral eigenspaces.

For fixed neutral geometry, repeated Schur reclosure suppresses cross-eigenspace couplings and converges to spectral pinching onto the commutant of Y . Accordingly, long-run universal emergence cannot be identified with endless repetition of a frozen geometry; the full BFG orbit must be allowed to reconstruct its own state data [2].

Synthetic operator audit: noncommutativity and spectral reformation

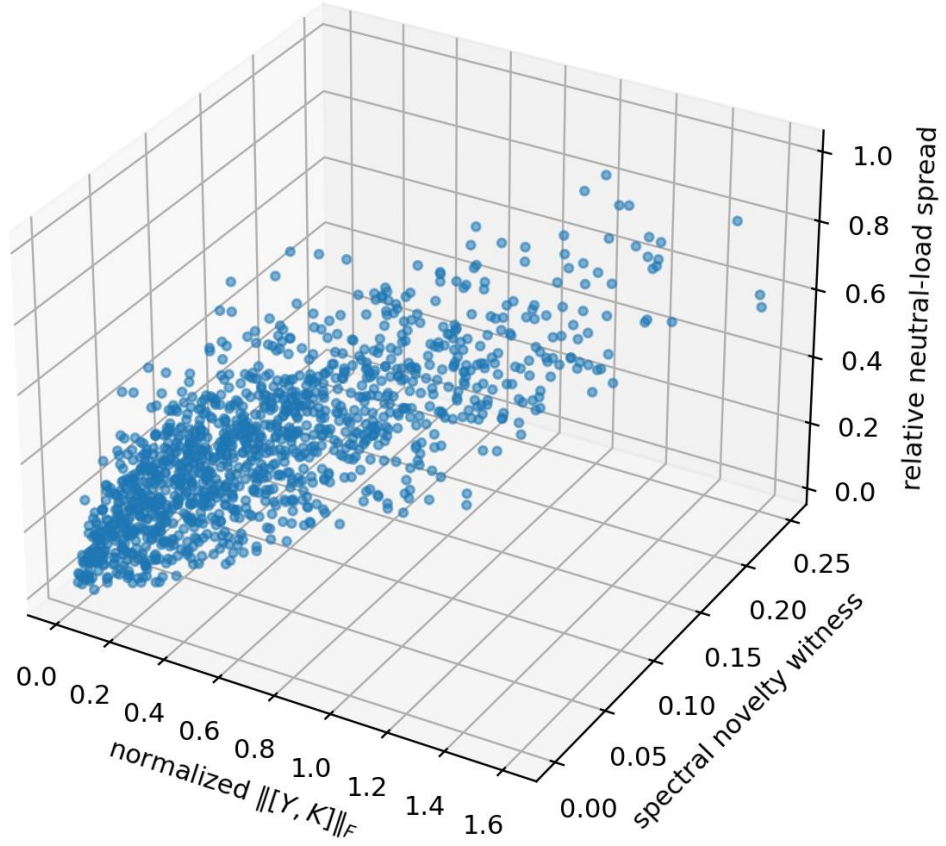


Figure 5. Synthetic 3×3 operator audit (1,300 random trials; seed 20260926) illustrating the relationship among normalized commutator magnitude, spectral novelty, and neutral-load differentiation. This figure is a new mathematical visualization for this preprint and is not empirical evidence.

12. Dimension monotonicity, terminality, and boundary discipline

The active target carrier is the reduced range of the selected analysis map. Therefore finite reclosure cannot create unbounded carrier dimension from a finite-dimensional start.

$$\dim \mathcal{H}_{n+1} = \text{rank } \mathcal{A}_n \leq \dim \mathcal{H}_n \quad (41)$$

$$\dim \mathcal{H}_n \leq \dim \mathcal{H}_0 \quad (42)$$

Emergence is therefore structural reorganization within, or compression of, an available carrier; it is not purchased by arbitrarily appending new basis directions. Persistent rank can nevertheless decrease or increase inside the active carrier through exact loss of inherited support and the canonical emergent seed mechanism.

12.1 Exact terminal conditions

- No persistent source or no closure-positive selected source sector.
- Exact identity-witness annihilation.
- Exact formation annihilation after Selection.
- Exact vanishing of a required dual branch load.
- Exactly zero analysis rank.
- An emergent complement that lacks the required unique simple negative ground seed.

12.2 Numerical ambiguity is not terminality

Floating-point uncertainty near a rank boundary, $y=1$ Selection boundary, or very small positive dual load is reported as numerical ambiguity. No implementation tolerance is promoted to a physical constant. This distinction is explicit in the living package audit [3].

13. Computational and numerical audit status

The BFG source line contains several distinct audits. They test mathematical identities, implementation consistency, and discrimination of synthetic operator cases. They do not establish natural realization.

Table 4. Mathematical and implementation audit ledger.

Audit	Reported sample	Result / diagnostic	Interpretation boundary
Canonical universal reclosure consistency audit	10,000 complex random instances	Neutral partition, reciprocal balance, compact/expanded equivalence, polar support, positivity, non-expansion checked to floating precision.	Internal mathematical consistency only [1].
Canonical emergence audit	10,000 trials	9,993 formation-gate passes; all 9,993 successful trials had a nonzero spectral novelty witness; 1,000 commuting controls stayed at roundoff.	Synthetic discrimination test, not physical probability [1].
Analytic novelty falsification audit	20,000 random + 5,000 commuting + 5,000 ambient-support tests	Closed Schur formula and moment identity reproduced to floating precision.	The theorem is analytical; simulation is implementation stress [2].
Archived fundamental-closure regression audit	174 tests	PASS, Python compile PASS, package consistency PASS (25 Sep 2026).	Archived package status [3].
Working status supplied for this preprint	198 regression tests	Author-supplied 26 Sep 2026 working count.	Use the repository-certified count when this preprint is formally archived.

The differing regression counts in the final two rows are intentionally shown rather than silently reconciled: the retrievable 25 September package audit documents 174 passes, whereas the author's 26 September development-status note supplied for this preprint records 198. This is a provenance difference, not a mathematical result.

14. Continuum, memory, and phase

The committed finite state does not include continuum phase or full historical memory. Continuum research introduces closure-depth or recursive phase charts only after the finite event law is fixed [3]. On locally invertible charts, a previous anchor can be reconstructed from the current state; an explicit memory coordinate is required only when several mathematically distinct continuations share the same instantaneous readout.

$$\text{Memory} = \text{minimal branch information required for unique continuation} \quad (43)$$

This makes memory a continuation structure rather than a second copy of the entire historical state. The normalized finite witness ρ_W still carries current identity continuity across discrete reclosure events.

14.1 Phase and persistent transport

The canonical finite recursion representative places eigenvalue 1 on the persistent block and is sufficient for finite stability bookkeeping. A richer continuous physical branch may retain nontrivial unit-modulus phase/holonomy. Such phase is not inserted into the finite minimal state merely because it may be needed later; it must arise from the continuum extension or a derived physical quotient [9,11–13].

15. Minimal polar carrier and the geometric branch

The BFG minimal-carrier program is downstream of the finite closure law. Under explicit branch conditions—two independent noncollapsed polarity roles, a nonzero relation, retained conjugate phase, and freedom-closed minimality—the minimal complex role carrier is $\mathbb{C}^2$ [9]. The real Hermitian readout then has four dimensions.

$$\mathcal{H}_{\text{pol}} \cong \mathbb{C}^2 \quad (44)$$

$$\text{span}_{\mathbb{R}}\{P_+, P_-, Q_1, Q_2\} = \text{Herm}(2, \mathbb{C}) \quad (45)$$

$$X = x^0 I + x^1 \sigma_1 + x^2 \sigma_2 + x^3 \sigma_3 \quad (46)$$

$$\det X = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2 \quad (47)$$

The determinant identity is standard two-spinor/Hermitian-matrix mathematics [31]. In BFG it is relevant only after the polar-conjugate branch conditions have been obtained internally; it is not used as a constitutive premise of the finite closure law.

Null-cone slice: $(x^0)^2 - (x^1)^2 - (x^2)^2 = 0$ at $x^3 = 0$

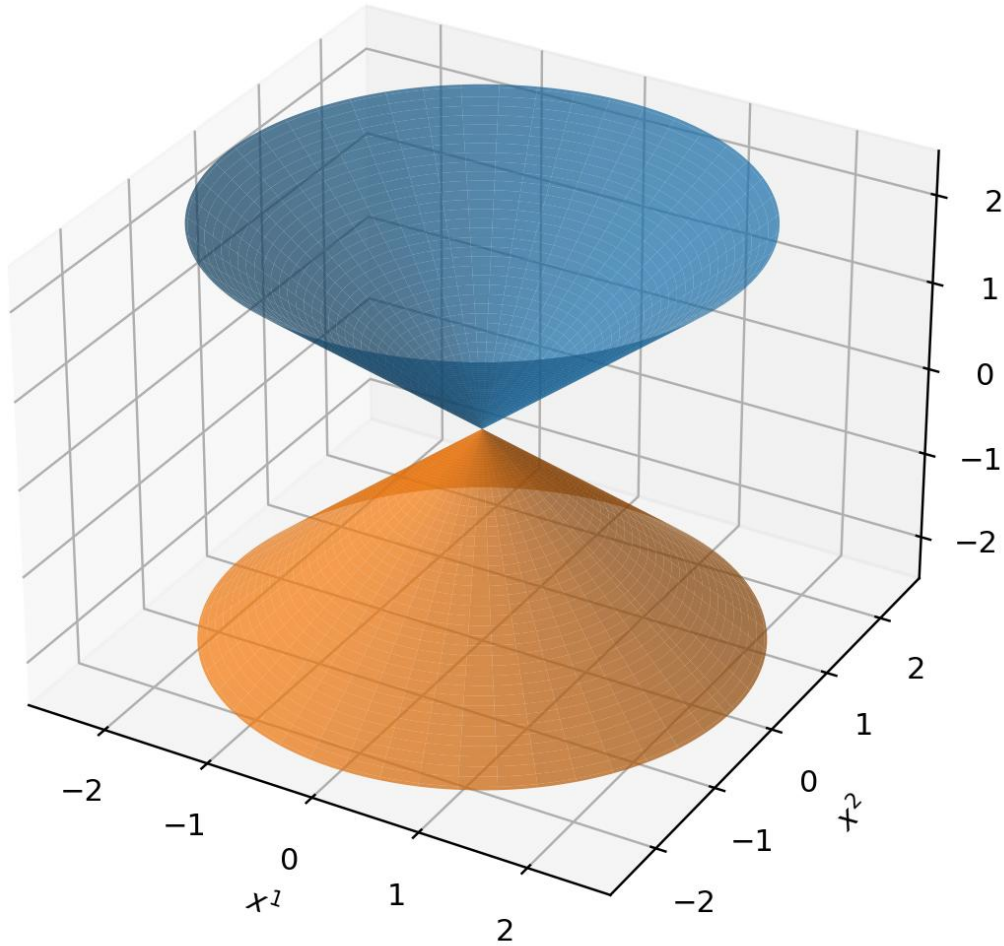


Figure 6. Three-dimensional $x^3=0$ slice of the null set $\det X=0$ for $X \in \text{Herm}(2, \mathbb{C})$. The double cone is a downstream geometric representation of the minimal polar-conjugate branch, not a primitive BFG spacetime assumption.

15.1 Universal action-scale branch

For a continuous persistent phase representation $U(\tau)$ with additive phase frequency ω , the BFG action-scale closure imposes continuity, additivity, positive orientation, and no-retuning on the phase-to-energy readout. Under these assumptions the readout is linear [9].

$$E(\omega) = S_0 \omega \quad (48)$$

An isolated first positive persistent phase rate Ω_0 then defines internal temporal and energetic scales. This is a branch theorem, not part of the five-object finite state.

16. Relation to the rich BFG semantic state

The reduction to five finite roles does not erase the older BFG state language. It organizes it. A richer source-facing state may retain the carrier, the difference/coherence/neutral-admissibility capacity triple, and the detailed recursion operator:

$$\Xi_{\text{rich}} = (\mathcal{H}, \mathcal{D}, c, \mathfrak{N}, \rho_F, \rho_W, Y, R_C) \quad (49)$$

$$K = c + \mathfrak{N} - \mathcal{D}, \quad P = P_{\text{per}}(R_C) \quad (50)$$

The rich state is therefore a semantic/source lift when separate capacity meanings, recursive phase, or domain-level readouts are required. The finite quotient asserts only that these details are unnecessary for computing the current finite successor after K and P are known [3,4].

Likewise, earlier coarse-grained BFG variables such as order, information, attractor organization, temporal continuity, self-consistency, memory, and closure stability may be retained as downstream observables or semantic coordinates. They should not be reintroduced as independent finite generator variables unless an explicit counterexample shows that the five-object state loses necessary information.

17. Falsification surfaces and open proof obligations

Table 5. Principal open proof and realization obligations.

Obligation	Current status	Failure meaning
Neutral Selection source synthesis	Parameter-free and adopted in the living finite law; cross-source rather than one historical equation.	If a deeper BFG derivation gives a different universal retain/export boundary, the present finite selector must be revised.
Noncommuting same-rank continuum	Open.	Finite event law would not yet determine a unique local continuous interpolation.
Persistent phase / memory reduction	Partly closed on local invertible charts; globally branch-dependent.	Additional continuation data may remain necessary.
Minimal polar carrier	Closed only on the declared noncollapse/nonfragmentation/conjugate-phase minimal branch.	Failure of a branch gate terminates that geometric derivation.
Gravity / Einstein branch	BFG-internal conditional reduction exists, but universality from the frozen master law remains a proof obligation [10].	GR-like recovery would remain branch-specific or incomplete.
Gauge and matter structure	Open at absolute-master level.	Would require imported structure or leave sector freedom.
Quantum statistics / measurement	Open.	A universal quantum recovery cannot yet be claimed.
Cosmology and dimensionless predictions	Open.	No completed ToE-level empirical closure.

Obligation	Current status	Failure meaning
Independent empirical realization	Open.	Mathematical closure would not imply natural truth.

These obligations are deliberately not filled by importing external physical theories. External references [32 – 38] provide standards against which later BFG recovery claims must be compared; they do not define the finite BFG generator.

18. Conclusion

The current finite BFG program can be expressed more compactly than its historical source architecture without weakening its finite predictive closure. The central result is not a single giant formula, but a disciplined separation between one minimal typed state and one universal reclosure law.

$$\Xi_{\min} = (K, \rho_F, \rho_W, Y, P) \quad (51)$$

$$\Xi_{n+1} = \mathcal{U}_{\text{BFG}}(\Xi_n) \quad (52)$$

Inside $\mathcal{U}_{\text{BFG}}$, exact neutral resolution determines the complementary responses; Neutral-Contrast Selection removes the need for a fitted retain/export threshold; the selected formation density fixes reciprocal dual order; the resulting analysis map induces its own positive successor load; one common active frame transports formation; a separate polar overlap transports inherited identity; and a simple negative ground mode is the only source of genuinely new formation in the non-inherited target complement. Emergence is therefore a possible result of the same reclosure, not an additional fundamental force.

The finite architecture is now simple enough to state in one line yet rich enough to distinguish formation, identity, neutral mediation, persistence, reproduction, export, and structural emergence. The next development stage is consequently derivational rather than additive: extend the frozen master law into a local noncommuting continuum, derive phase and geometry without retuning, and test whether the same law recovers physical sectors and independent observations.

**Minimal state → neutral reclosure → induced geometry → inheritance or
emergence → reclosure**

Appendix A. Symbol register

Table 6. Core symbol register.

Symbol	Meaning
$\Xi_{\min}$	Minimal finite BFG state $(K, \rho_F, \rho_W, Y, P)$.
K	Hermitian formation/balance operator.
ρ_F	Positive nonzero formation/load density.
ρ_W	Normalized positive identity witness.
Y	Positive neutral/master load.
P	Orthogonal persistent projector.
C_N	Retained neutral response $(I+Y)^{-1}$.
B_N	Complementary mediated/export response $Y(I+Y)^{-1}$.
G	Graph metric $I+Y$.
Z	Neutral contrast $C_N - B_N$.
Q_N	Closure-positive Neutral-Contrast selector.
T_{sel}	Polar survival transport from old persistence to selected source.
Λ_C, Λ_B	Selected persistent neutral closure loads.
α, β	Reciprocal dual weights.
$\mathcal{A}$	Selected cross-fed analysis map.
U, Σ, V	Reduced SVD factors of $\mathcal{A}$.
T_{R4}	Polar inherited-support overlap transport into active target.
Π_E	Projector of the unique simple negative emergent ground mode.
$\perp$	Absorbing terminal/non-generation state.

Appendix B. Reproducibility and figure-generation note

Figures 2–4 and 6 are direct visualizations of equations displayed in this preprint. Figure 5 is a synthetic complex-Hermitian operator audit generated with NumPy using seed 20260926. Each trial draws a 3×3 Hermitian formation operator, a positive diagonal neutral load, and $\alpha \in (0.05, 0.95)$, then applies the exact Schur multiplier of Equation (45). The figure is included only to visualize the analytical theorem; the theorem does not depend on the simulation.

All numerical counts imported from BFG source documents are labelled as source-reported audits. No figure in this preprint uses observational data.

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Declarations

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